

Controlled Evidence for Retrieval-Signal Degradation as Context Fills: A Mechanistic Case Study in Gemma-2-2B-it

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Abstract

Long-context language models lose access to stored facts well before their context window is exhausted (“context rot”), yet the underlying mechanism has been reported only correlationally. Using **Gemma-2-2B-it** within its native 8k window, we build *controlled haystacks* from LongMemEval-S material: the same question is evaluated at every fill level $k \in \{0, 1, 2, 4, 8, 14\}$ distractor sessions, with the gold session’s position randomized and logged. On 140 questions, accuracy declines overall from 0.71 at $k = 0$ to 0.34 at $k = 14$ (the decline is not strictly monotonic; $k=4$ slightly exceeds $k=2$). Retrieval-head attention mass on the gold span and recall-associated sparse-autoencoder (SAE) feature activations both decline with k (mixed-effects $\beta_k = -0.023$, $p=2.4 \times 10^{-191}$; feature $\beta_k = -0.535$, $p=3.0 \times 10^{-90}$), controlling for gold position. In a *paired within-question* design the fill level at which the internal signal collapses is associated with the level at which the answer flips (Spearman $\rho = 0.731$, $p=3.6 \times 10^{-9}$, $n=48$). Intervention experiments give *suggestive* causal evidence: mean-ablating the 20 candidate retrieval heads at $k=0$ drops accuracy by 0.82 (single random-head control: 0.25), and feature-mode re-injection at high k recovers 0.161 (95% CI [0.032, 0.290]) of 31 flipped answers. We deliberately frame this as a single-model, single-seed *case study*: the SAE features are selected in-sample, the attention measure is unnormalized, and Gemma-2’s sliding-window layers are a partial confound — all addressed as open issues (§4) rather than resolved here. The pipeline is seeded and re-runnable byte-identically.

1 Introduction

Context rot — the degradation of retrieval as the context fills, well inside the model’s nominal window — has been documented behaviorally but not mechanistically. Two interpretability instruments let us watch the failure as it happens: *retrieval heads*, attention heads that copy from the evidence span into the generated answer, and *SAE features*, sparse directions that activate when the model recalls a fact. We ask three questions:

- H1** Does the internal retrieval machinery (attention mass on gold, recall-feature activation) degrade as context fills?
- H2** Does the *per-question* collapse point of that internal signal predict the *per-question* behavioral flip point?
- H3** Is the machinery causally responsible for the correct answer?

The novelty over prior correlational context-rot reports is a paired within-question design in which each question is its own control, isolating the effect of context fill from the effect of question difficulty.

2 Method: controlled-haystack paired design

Figure 1 summarizes the pipeline. From LongMemEval-S we keep the 140 questions that (i) are non-abstention, (ii) have *exactly one* gold evidence session of $\leq 2,000$ tokens (long gold sessions are trimmed to a turn window around the `has_answer` evidence turns), and (iii) are not of type **single-session-preference**, whose gold “answers” are prose preference-rubrics that the heuristic substring/F1 grader cannot score.

For each question and each k we pack the gold session plus k topic-matched distractor sessions (each truncated to a fixed per-distractor cap so context grows roughly linearly with k) into $\leq 7,500$ tokens, placing the gold session at a uniformly random, logged position. A question enters the H2/H3 analysis only if it is answered correctly at $k=0$, so any failure at $k > 0$ is attributable to context fill, not missing knowledge.

We instrument Gemma-2-2B-it with eager attention. Retrieval heads are identified by copy-score probes built from benchmark vocabulary (20 heads selected); the five SAE layers $\{20, 21, 22, 23, 24\}$ are chosen by an empirical separability scan over all 26

Table 1: Accuracy by fill level k on 140 questions. Accuracy declines overall as distractor sessions are added — though not strictly monotonically ($k=4$ slightly exceeds $k=2$) — despite the gold evidence remaining present and in-window.

k	0	1	2	4	8	14
Acc.	0.714	0.671	0.579	0.586	0.521	0.343

Gemma Scope **res** layers. Decoding is greedy; seed 0; CUBLAS_WORKSPACE_CONFIG=:4096:8 with deterministic algorithms makes a re-executed run byte-identical on `answers.parquet`.

3 Results

H1 (descriptive). Retrieval-head attention mass on the gold span declines with k ($\beta_k = -0.023$, $p = 2.4 \times 10^{-191}$; mixed-effects model with question as a random intercept and gold position as a covariate, gold-position $\beta = 0.446$). Recall-feature activation shows the same decline ($\beta_k = -0.535$, $p = 3.0 \times 10^{-90}$). See Figures 2 and 5.

H2 (paired). Among the 100 questions correct at $k = 0$, the per-question signal-collapse level is associated with the answer-flip level: Spearman $\rho = 0.731$ ($p = 3.6 \times 10^{-9}$, $n = 48$ questions with both events observed); the feature-signal variant gives $\rho = 0.763$. We caution that both quantities are event-times over the same six k levels, so a global worsening trend can inflate this correlation in the absence of a per-question link; a permutation test that preserves the marginal k trend, threshold-sensitivity sweeps, and bootstrap CIs are required to separate the two and are not yet run (§4). We previously reported a Cox hazard ratio here; we withdraw it pending a discrete-time hazard model appropriate to the irregular, six-level k grid. See Figure 3.

H3 (intervention). Mean-ablating the 20 candidate retrieval heads at $k = 0$ on the 100 previously-correct questions drops accuracy by 0.82, versus 0.25 for a single matched random-head set. This shows the heads matter for answering at $k = 0$; it does *not* by itself establish that they cause the high- k failure. Intervening at high k in feature mode (SAE recall-feature re-injection) recovers a modest 0.161 (95% CI [0.032, 0.290]) of the 31 flipped answers. Because the features are selected in-sample and only one random control is run, we read H3 as suggestive rather than dispositive; stronger controls (many random head sets, layer- and magnitude-matched heads, sham/anti-recall feature injection) are listed in §4. See Figure 4.

4 Limitations and open issues

We state these prominently because several are potential confounds for the headline claims, not merely scope caveats. They define our planned next round.

In-sample SAE feature selection (label leakage).

Recall features are chosen by correlation with correctness on the same (q, k) samples on which H1/H2/H3 feature results are then reported. This is circular. The fix is a discovery/held-out question split with feature and layer selection on discovery only, plus random-feature, anti-correlated-feature, and magnitude-matched controls; the feature results should be read as in-sample proxies until then.

Unnormalized attention mass (dilution).

We report absolute retrieval-head attention mass on the gold span. As k grows the key set grows, so mass on any fixed span can fall by dilution alone. Required controls: mass normalized by the gold-token fraction, enrichment over uniform, and length/position-matched distractor- and random-span baselines.

Sliding-window visibility as a competing mechanism.

Gemma-2 interleaves sliding-window (4,096) and global layers. With gold position randomized and context growing, high- k conditions place gold outside the sliding window for many layers, so part of the attention decline may be architectural rather than emergent. We must stratify by in/out-of-window gold, analyze global layers separately, and add a gold-stays-visible condition.

H2 shared-trend artifact and Cox fragility.

Collapse- and flip-times live on the same six-level k grid; a global trend can induce correlation without a per-question link. We need a permutation test preserving the marginal k trend, threshold sweeps (collapse_frac $\in [0.3, 0.7]$), bootstrap CIs, and a report of how often collapse *precedes* flip. The previously reported Cox HR is withdrawn pending a discrete-time hazard model with only prior/current k signal.

Causal strength.

H3a shows the heads matter at $k = 0$, not that they cause high- k rot; H3b recovery is modest and rests on in-sample features. Stronger controls (many random head sets, layer-/magnitude-matched heads, sham and anti-recall injection, harm checks on already-correct high- k items) are needed before any causal language.

Heuristic grader.

Substring/token-F1 correctness drives accuracy, feature selection, flip detection, and recovery detection. We will audit a stratified sample (cor-

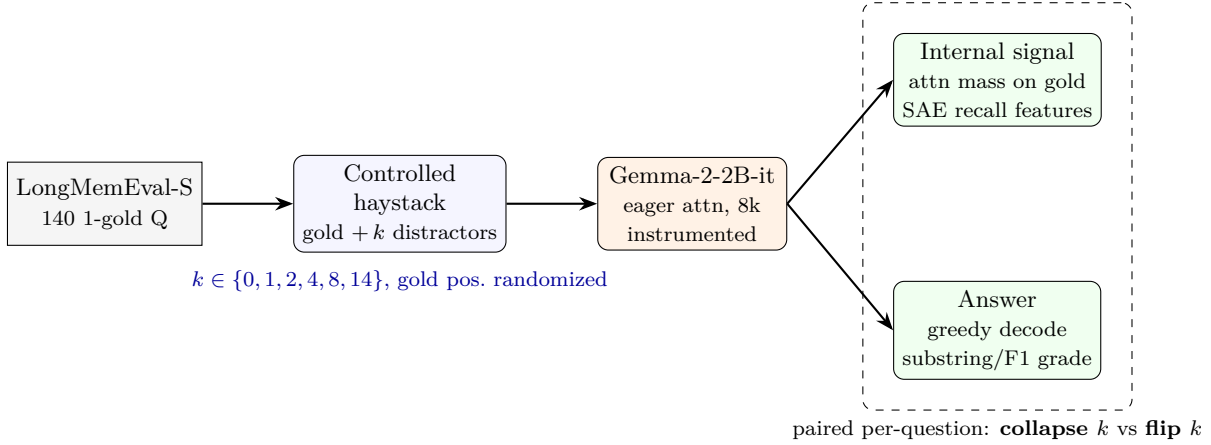


Figure 1: Controlled-haystack paired pipeline. Each question is replayed at six fill levels with the gold session at a randomized position; an instrumented forward pass records both the internal retrieval signal and the graded answer, which are aligned per-question to test whether internal collapse predicts the behavioral flip (H2) and whether the machinery is causal (H3).

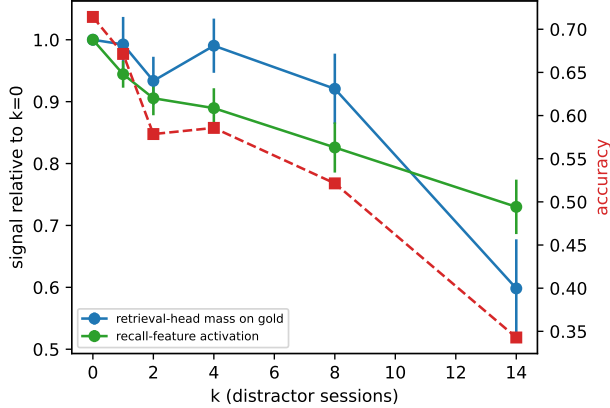


Figure 2: H1. Internal retrieval signal (attention mass on the gold span) versus fill level k , with question as random intercept. The signal decays as context fills even though the gold evidence is present and in-window.

rect/wrong/flipped/recovered) against human or LLM-judge labels, report grader precision/recall, and re-run headline stats on the H2/H3-critical examples.

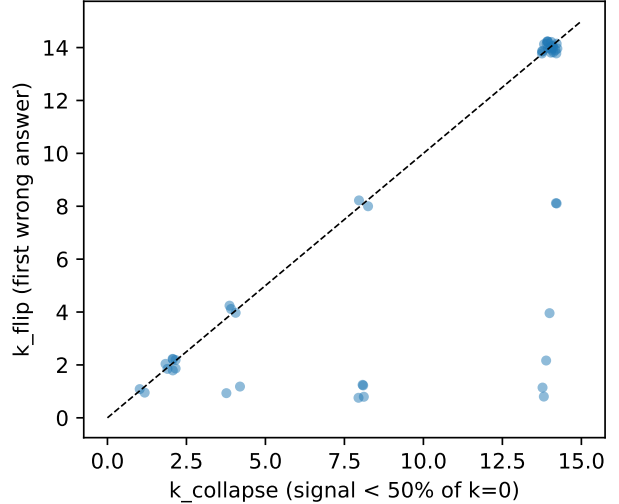


Figure 3: H2 (paired). Per-question internal-collapse level versus behavioral flip level. The two are associated ($\rho = 0.731$); because both are event-times over the same six k levels, a permutation test preserving the marginal k trend is needed before reading this as a per-question link.

5 Related work

Benchmark variant and sampling. Long gold sessions are trimmed to a turn window around the `has_answer` evidence ($\approx X\%$ of questions; to be reported), making this a controlled LongMemEval-S *variant*; we will compare trimmed vs. untrimmed behavior. Gold position is sampled once per (q, k) ; given the large position effect ($\beta = 0.446$), multiple position seeds are needed. Results are single-model (Gemma-2-2B-it) and single-seed; eager attention is required for the hooks.

Behavioral context degradation. “Lost in the middle” (Liu et al., 2023) shows accuracy depends on evidence position within long contexts; “context rot” reports (Chroma, 2024) document accuracy decay as inputs lengthen. These are behavioral and largely correlational; our contribution is a paired, evidence-in-window design that pairs the behavior with internal signals and intervention.

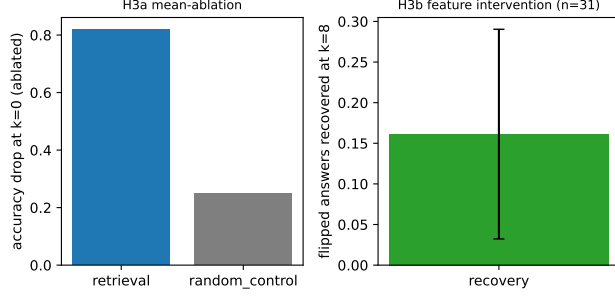


Figure 4: H3 (intervention). Left: accuracy drop from mean-ablating the 20 candidate retrieval heads (0.82) versus a single matched random-head set (0.25). Right: fraction of flipped answers recovered by feature-mode intervention (0.161, 95% CI [0.032, 0.290]). Read as suggestive: features are selected in-sample and only one random control is run.

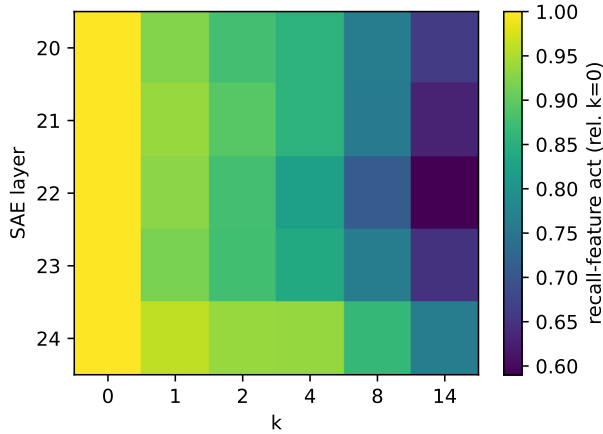


Figure 5: H1 (features). Recall-associated SAE feature activation across layers {20–24} as a function of k ; activation weakens with fill, mirroring the attention-mass decline ($\beta_k = -0.535$). Features are selected in-sample by correlation with correctness, so this panel is descriptive, not held-out evidence (§4).

Retrieval heads and mechanistic interpretability.

Retrieval heads (Wu et al., 2024a) are attention heads that copy salient context into generation; induction-head and circuit analyses motivate head-level ablation as a causal probe. We use copy-score head identification but caution (§4) that head necessity at $k=0$ is not sufficiency for high- k failure.

Sparse autoencoders. Gemma Scope (Lieberum et al., 2024) provides JumpReLU SAEs across Gemma-2 layers; SAE features have been used to read and steer model internals. Our use is intervention-based but, as noted, currently in-sample.

Long-term memory evaluation. LongMemEval (Wu et al., 2024b) supplies the question/evidence material; we adapt it into controlled haystacks rather than evaluating end-to-end memory systems.

6 Reproducibility

Seed 0; artifact schema v2; environment hash d6e719d3. The producing run is git 54907d9 plus the smoke-gate config/ data-prep amendment now committed as 83a762c (model $\rightarrow$ gemma-2-2b-it; exclude single-session-preference; $n=140$). Three SLURM jobs reproduce everything end-to-end; Run 1 re-execution is byte-identical on answers.parquet. Small summary artifacts that regenerate every headline number (run1_verify.json, results_summary.json) are tracked in the repository under results/2b/.

References

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